Lecture 8

Organelles of the plant cell and organisation into tissues

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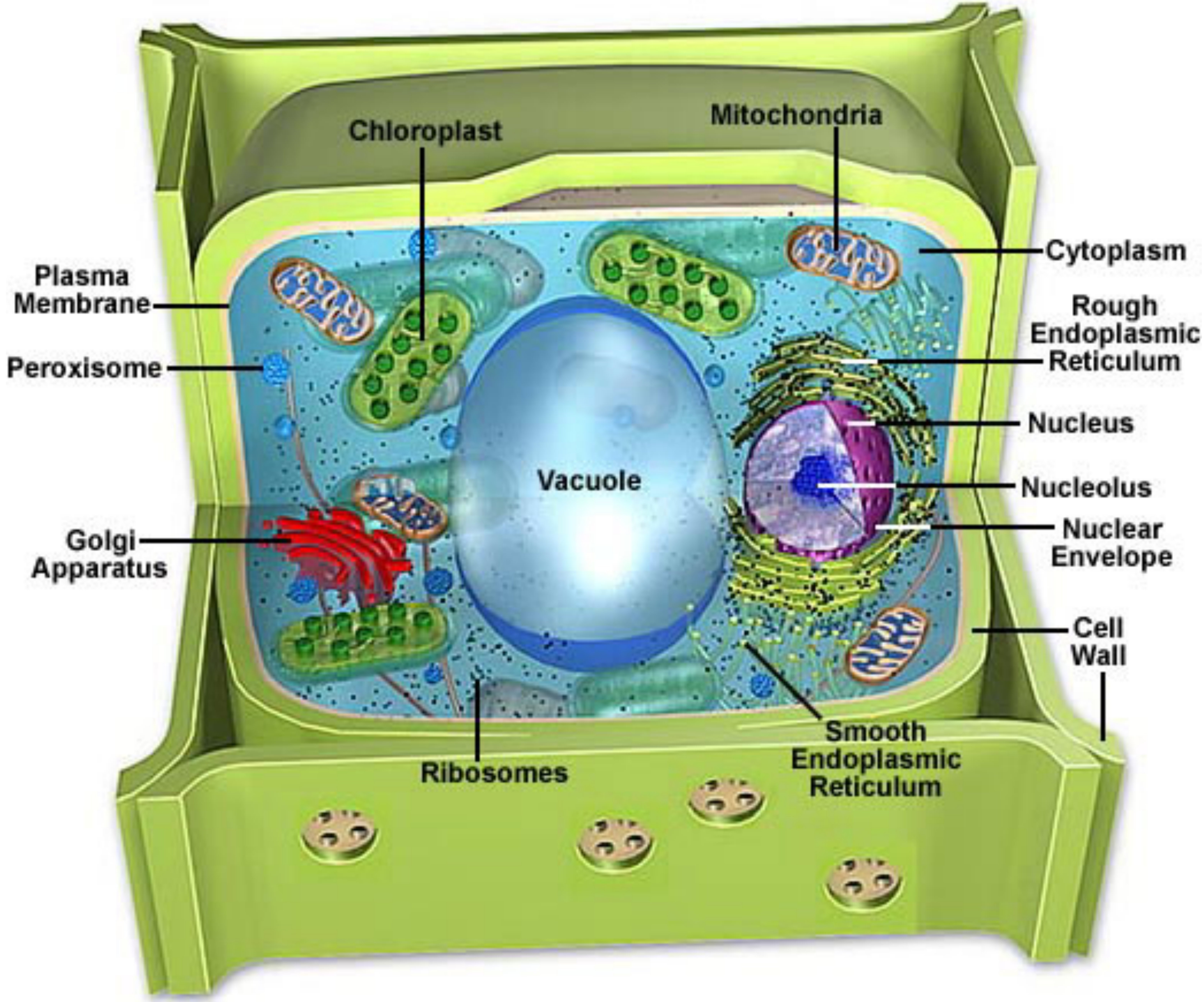
FOOD MICROBIOLOGY ACADEMY


Chloroplast
Mitochondria
Cytoplasm
Plasma Membrane
Rough Endoplasmic Reticulum
Peroxisome
Nucleus
Vacuole
Nucleolus
Golgi Apparatus
Nuclear Envelope
Cell Wall
Smooth Endoplasmic Reticulum
Ribosomes

Plant Cell Chloroplast Structure

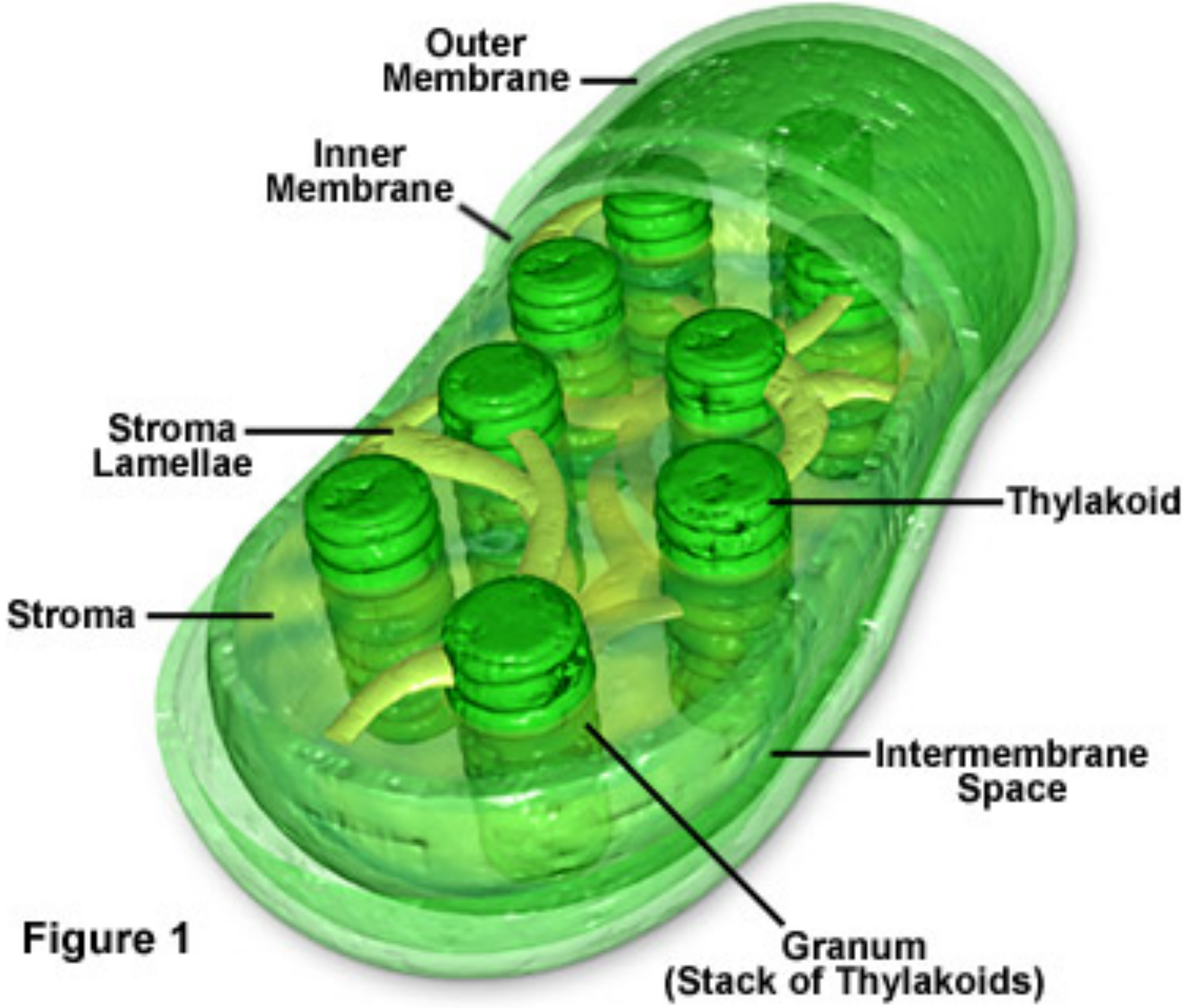
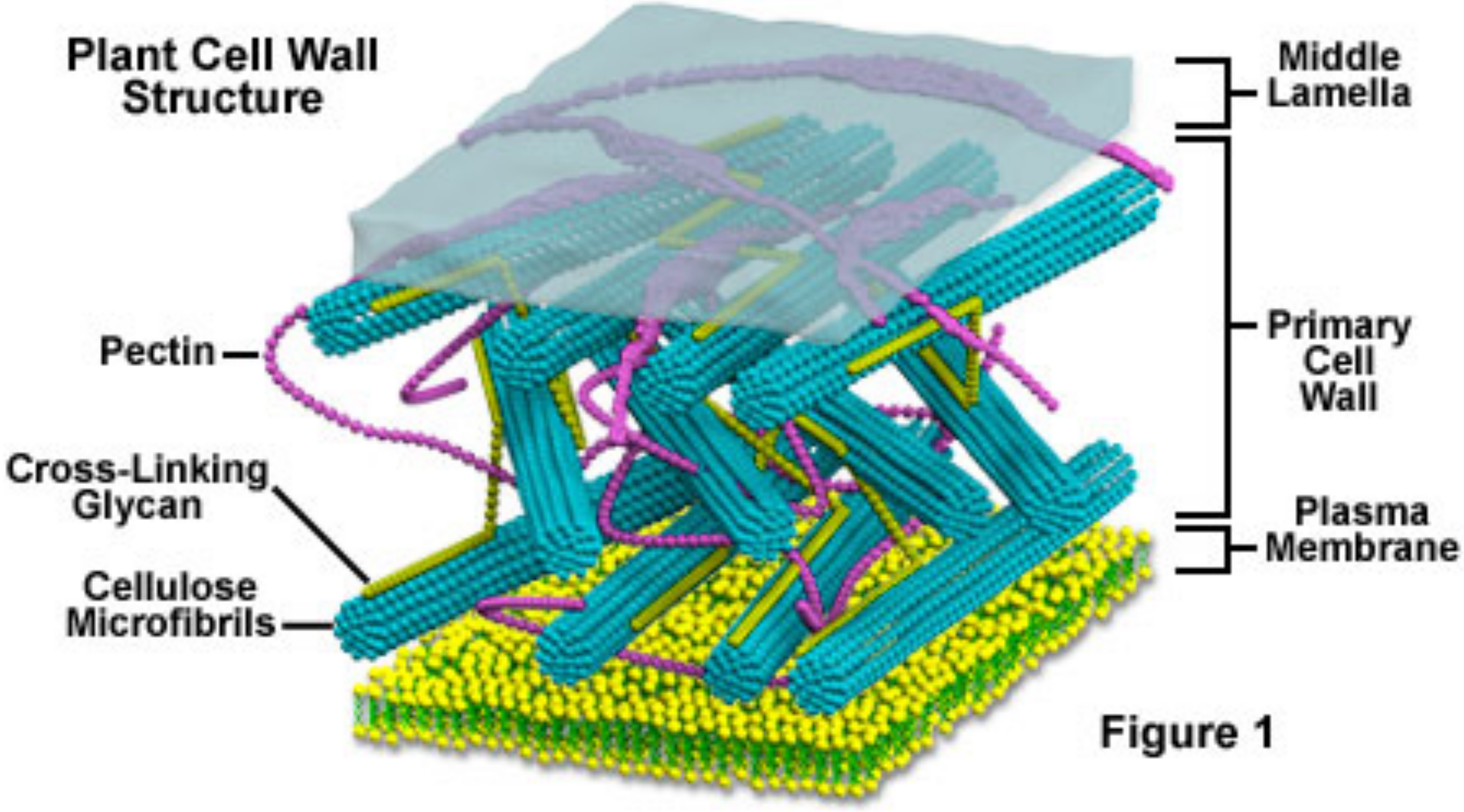
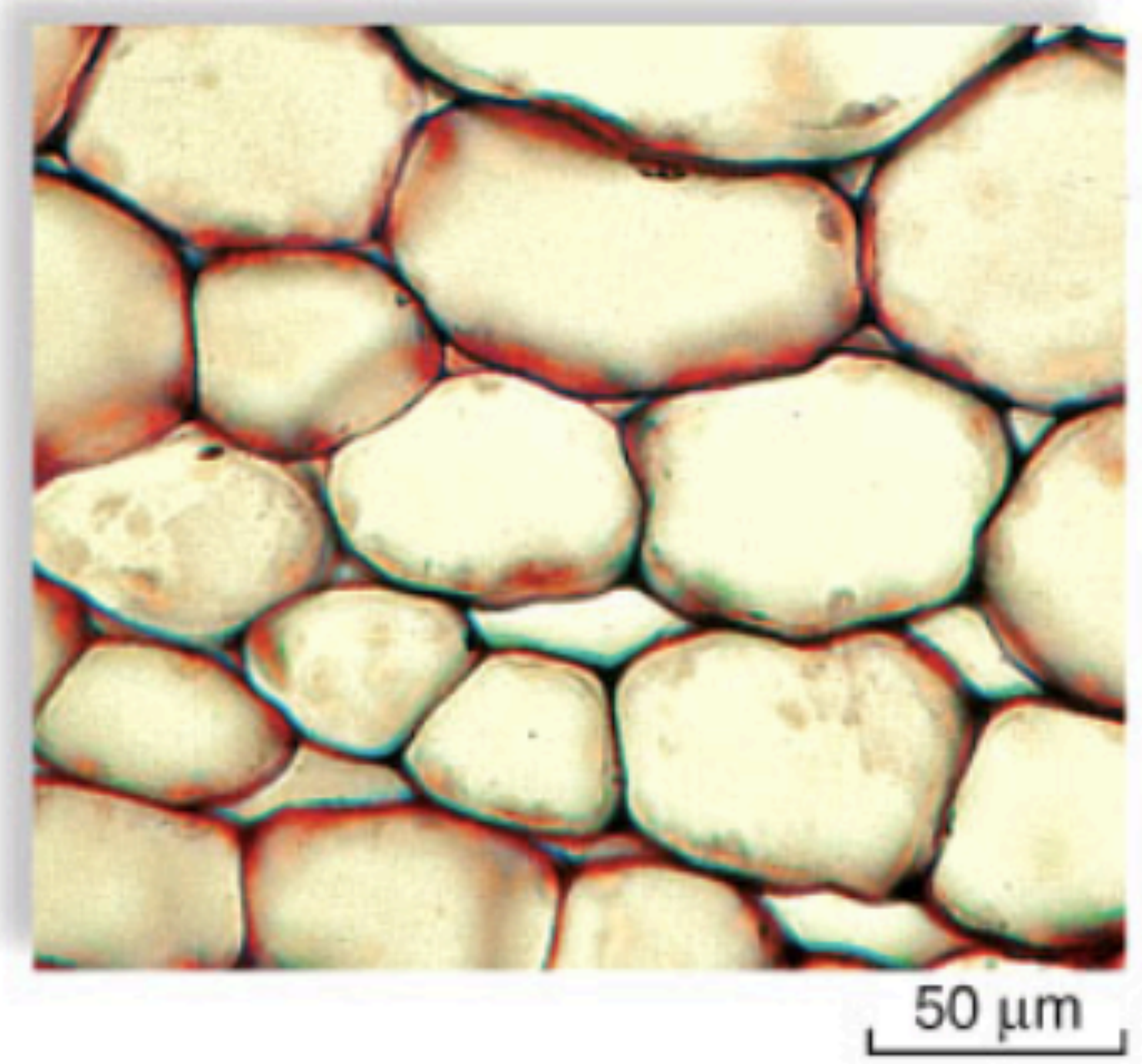


Figure 1


Plant Cell Wall
Structure
Middle
Lamella
Pectin
Primary
Cell
Wall
Cross-Linking
Glycan
Plasma
Membrane
Cellulose
Microfibrils
Figure 1

Parenchyma

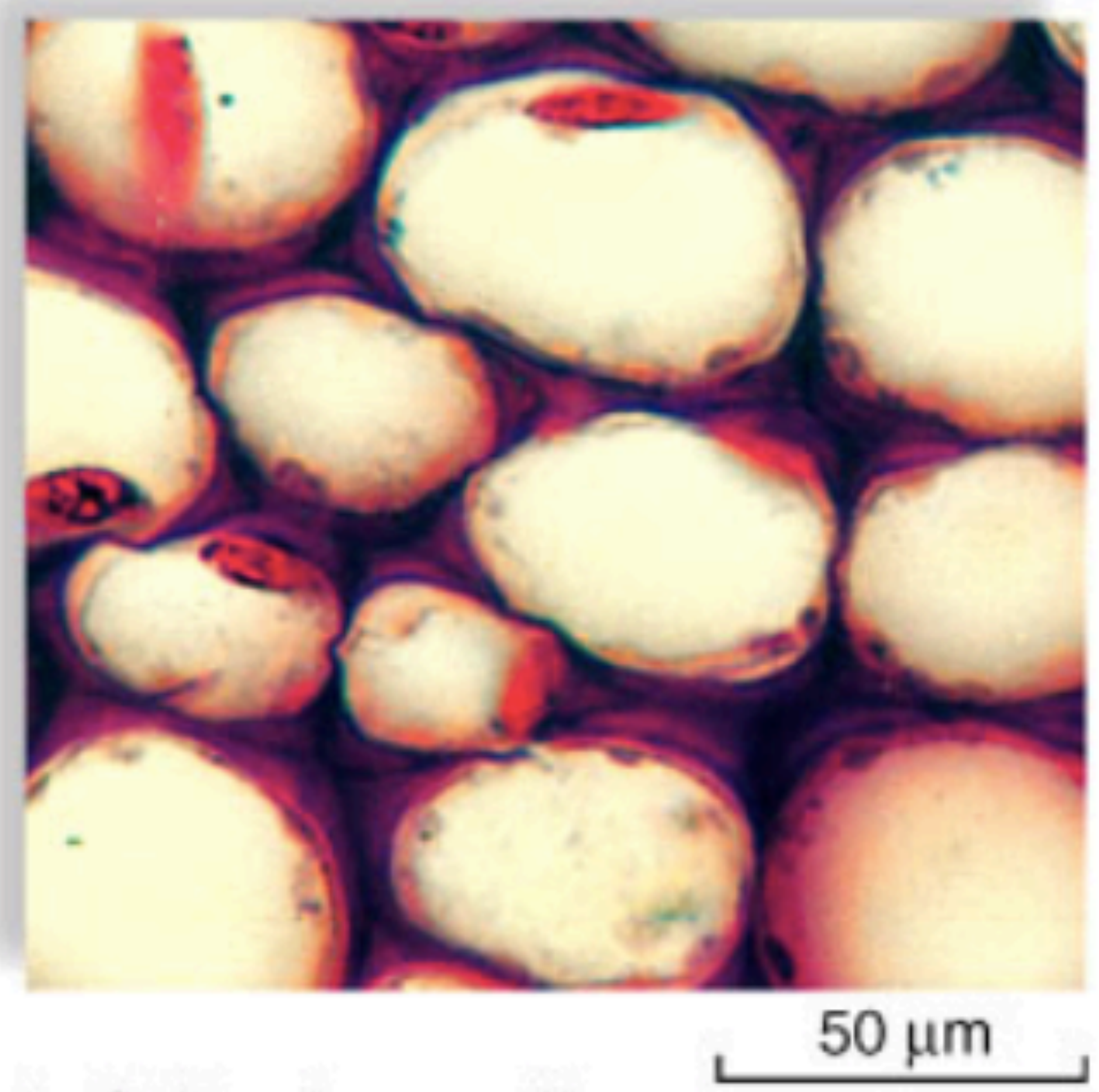
- Continuous cell masses
- Living at maturity
- Widespread
  - Cortex of stems/roots
  - Pith of stems
  - Leaf mesophyll
  - Flesh of fruits



a. Parenchyma cells

Collenchyma

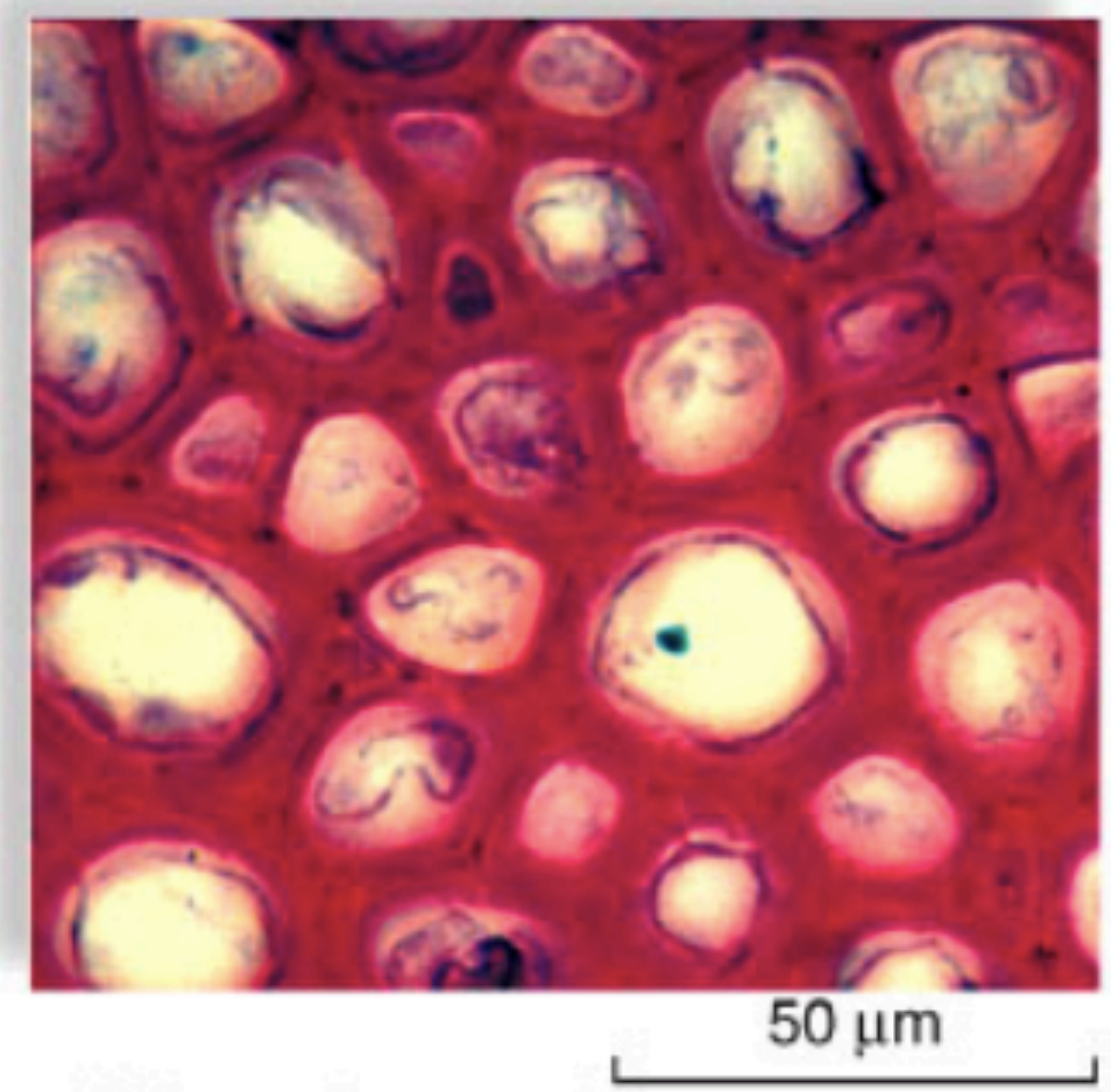
- Elongated cells
  - Support of growing organs
- Living at maturity
- Common as discrete strands or as continuous cylinders
  - Stems and petioles
  - Border eudicot veins



b. Collenchyma cells

Sclerenchyma

- May develop anywhere
  - Secondary
  - Primary
- Lack protoplast at maturity
- Two types
  - Fibers
  - Sclereids
- Principle characteristic is:
  - Thick, lignified secondary walls
  - Important in strength



c. Sclerenchyma cells